

Behavioral Differences in Rapid Object Recognition between Humans and Machines

Thomas Klein, Wieland Brendel and Felix A. Wichmann

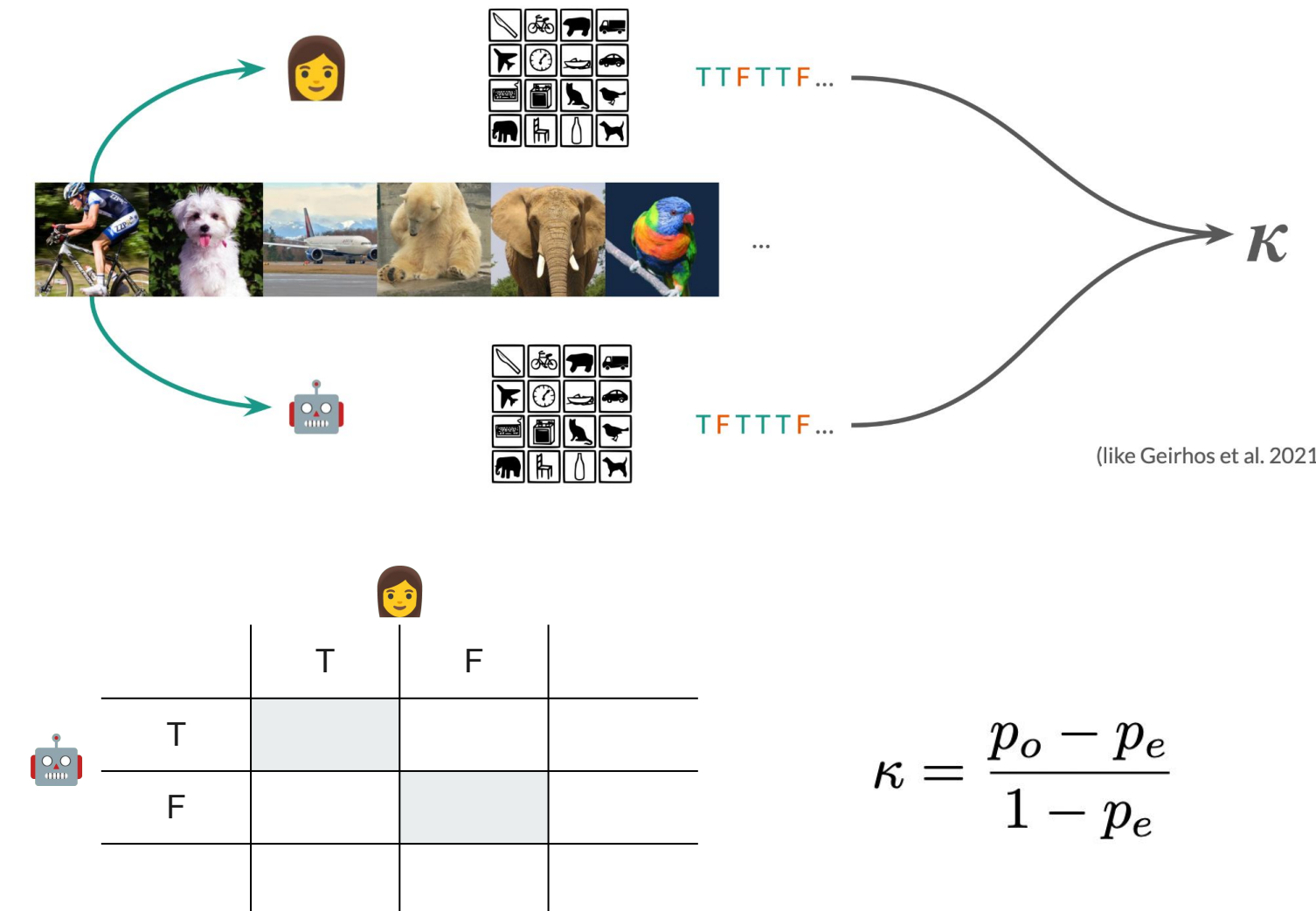
DNNs and human observers make different mistakes when classifying natural images.

Does this happen because humans have too much time to think about the images?

No! We find that even at very short presentation times, error consistency between humans and machines is very low.

Background

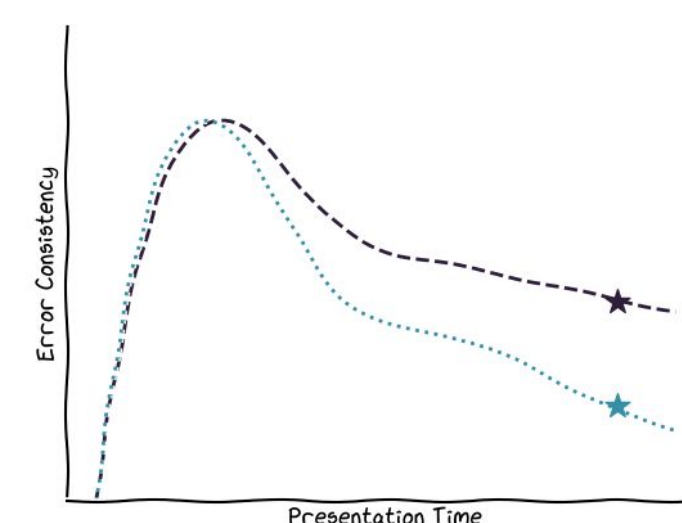
DNNs are currently considered the best models of the human visual system. But earlier work [1] finds that DNNs make different mistakes than humans, as quantified by *Error Consistency*:



Maybe, DNNs are good models of the ventral stream for the forward sweep, but then higher-level (recurrent) processing leads to behavioral differences.

Implication:

Higher EC at shorter presentation times.



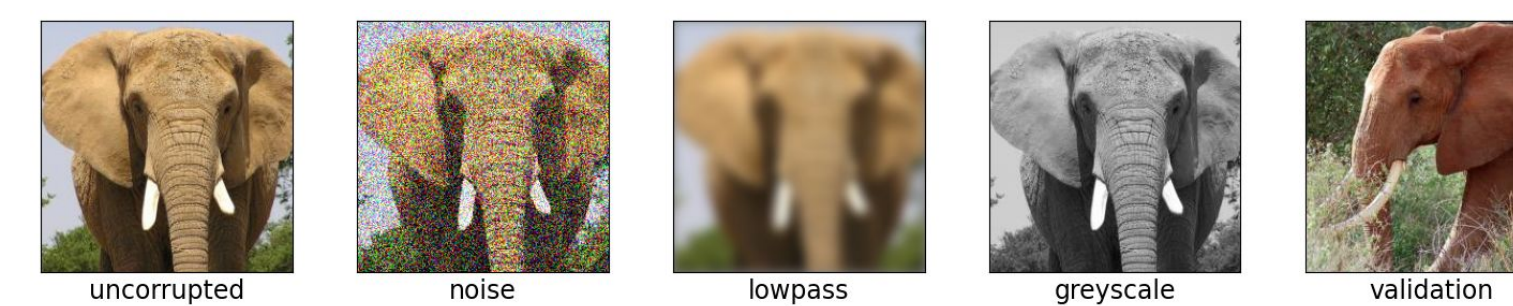
Acknowledgements

This work has been supported by the Machine Learning Cluster of Excellence, funded by EXC number 2064/1 – Project number 390727645. The authors would like to thank the International Max Planck Research School for Intelligent Systems (IMPRS-IS) for supporting Thomas Klein.

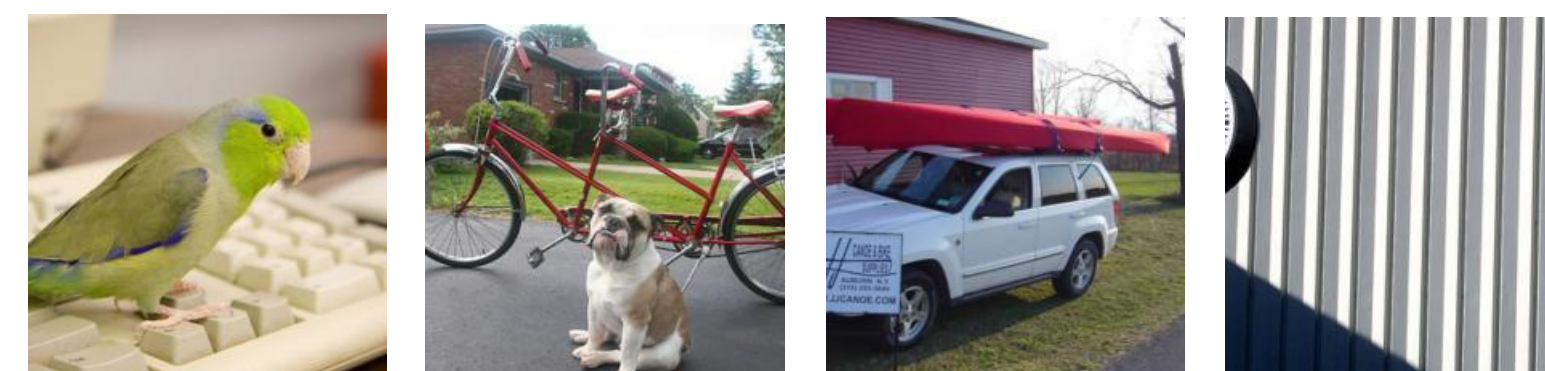
Experiment

We measure EC between humans and machines, but vary the presentation time from 8ms to 266ms.

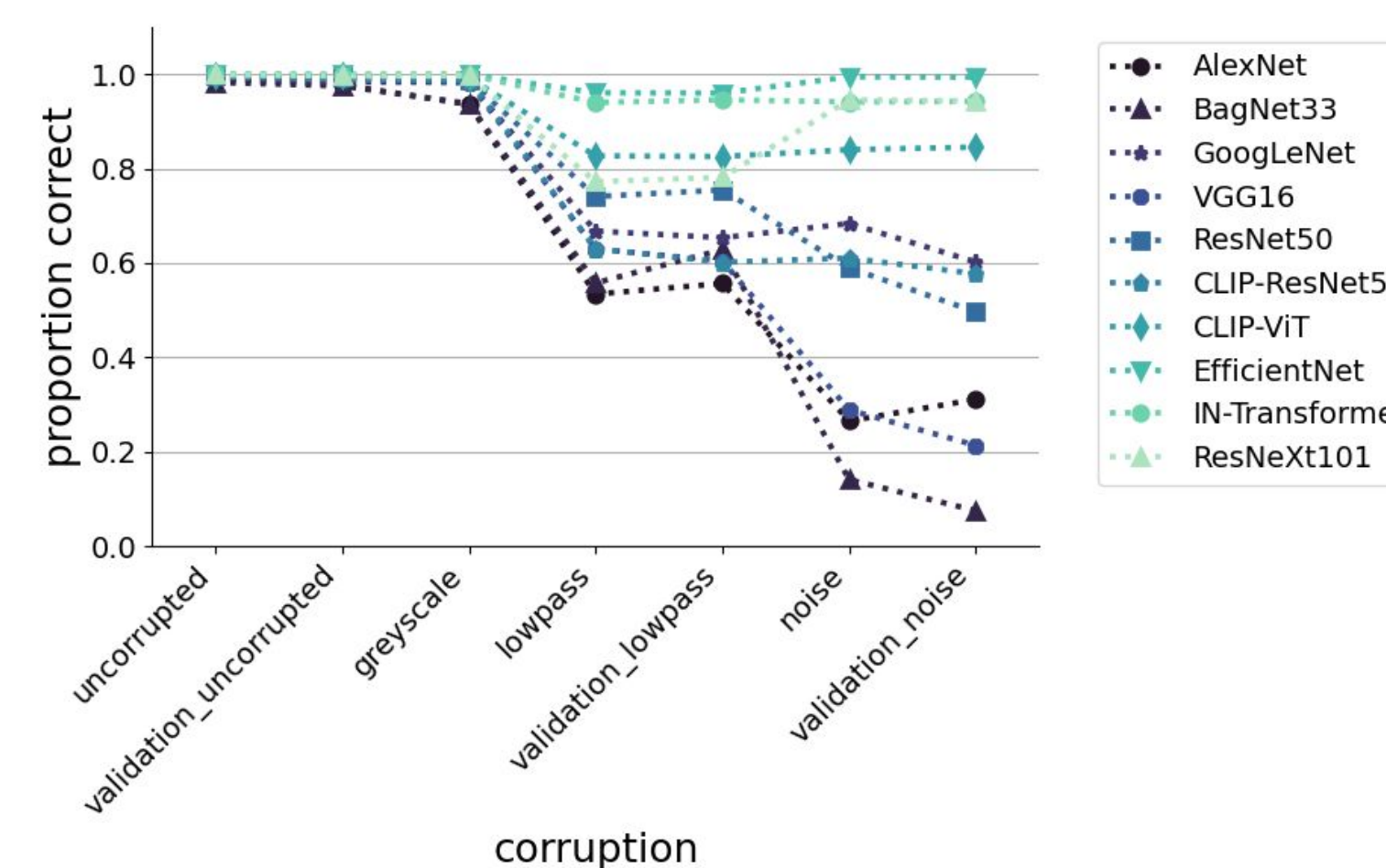
We apply corruptions to make images harder:



Unlike earlier work, we manually clean the images we use for our experiments, to avoid problematic stimuli:



We compare our human subjects to a suite of 10 DNNs (5 classic CNNs, 4 modern models)

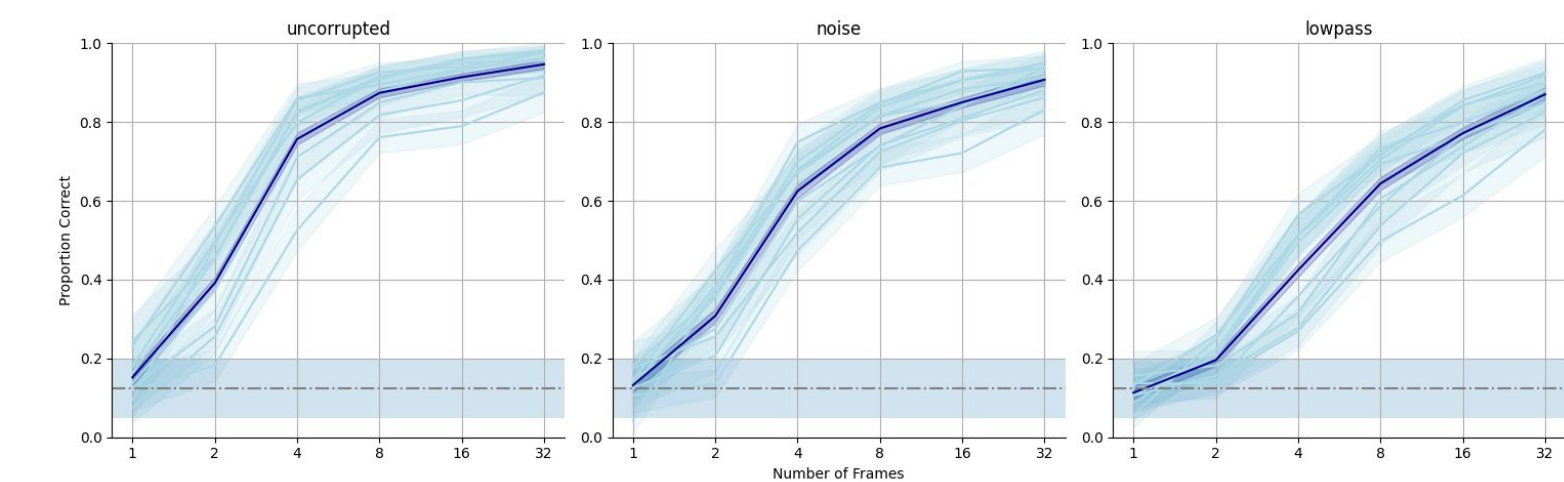


References

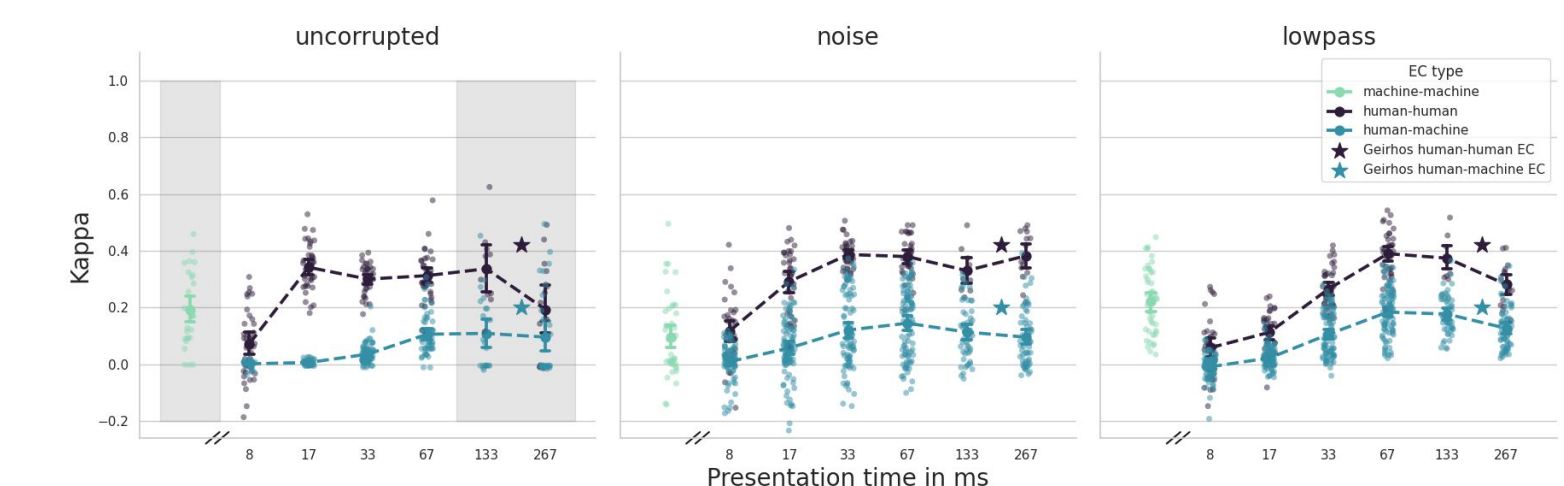
[1] Geirhos et al. (2021). Partial success in closing the gap between human and machine vision. *Advances in Neural Information Processing Systems*, 34, 23885-23899.

Results

1. Humans benefit from more presentation time, but can solve the task very early:



2. Across conditions, EC never exceeds values known from earlier work:



3. Error Consistency has a few issues:

- 😞 EC is undefined for two perfect raters
- 😞 EC is zero even for one perfect rater
- 😞 EC is agnostic to the number of classes
- 😞 EC is blind towards different types of mistakes
- 🧠 EC does not always lie in [-1,1]

😊 We are working on a tutorial paper that explains these issues, and will try to fix them!

In a hurry?
Find this poster
online!

